

Faculty of Engineering and Technology
B.Tech Programme
Semester VII

Deep Learning Laboratory

Course Code: 2604733 **Course Credit: 2**

PROGRAM OUTCOMES (POs):

PO 1 : The graduate will be able to apply the knowledge of mathematics, basic science, engineering fundamentals to solve the engineering problems.

PO 2 : The graduate will be able Identify, formulate and solve computer or IT related problems using acquired knowledge.

PO 3 : The graduate will be able to design software solutions or hardware components as per given specifications.

PO 4 : The graduate will be able to analyse, interpret, conduct investigations using engineering knowledge and provide valid conclusions.

PO 5 : The graduate will be able to use and demonstrate the use of the modern tools and technologies related to the latest trends in the computer industry.

PO 6 : The graduate will be able to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7 : The graduate will be able to understand the impact of the engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8 : The graduate will be able to apply ethical principles and commit to professional ethics and responsibilities.

PO 9 : The graduate will be able to function effectively as an individual, as a member or as a leader in a team.

PO 10 : The graduate will demonstrate competent Communication skills with respect to verbal communication and technical writing like effective reports, documentation and presentations.

PO 11 : The graduate will be able to demonstrate knowledge of Engineering and management principles and will be able to apply to manage projects in multidisciplinary environments.

PO 12 : The graduate will be able to recognize the need for, and will have the ability to engage in self learning as per real world requirements.

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PROGRAM SPECIFIC OUTCOMES (PSOs): **Track-3 (Artificial Intelligence and Machine Learning)**	
At the the end of the program the graduate would be able to:	
PSO 1:	Understand the fundamentals and underlying theoretical and practical concepts of different domains of computer science and engineering.
PSO 2:	Identify and analyze the problem, select the proper platform with appropriate data structure/database, storage and develop the solution with good algorithmic competence and proficiency in a wide range of programming languages.
PSO 3:	Develop and Implement the solutions for real world engineering problems using modern tools and techniques of artificial intelligence and machine learning.

Introduction:

The Deep Learning Laboratory course provides a hands-on environment to implement and experiment with core concepts and advanced techniques in deep learning. This lab emphasizes practical skills in designing, training, and optimizing neural networks using popular frameworks such as TensorFlow and PyTorch. Students will work on real-world problems, exploring applications like image recognition, natural language processing, and generative models. Through guided experiments and projects, learners will gain proficiency in configuring architectures, tuning hyperparameters, and deploying deep learning solutions, preparing them for research or industry roles in AI and machine learning.

Course Objectives:

- Gain practical experience in implementing and training deep learning models using popular frameworks like TensorFlow.
- Explore real-world applications of deep learning in fields such as computer vision, natural language processing, and generative models.
- Develop skills in optimizing deep learning architectures, including hyperparameter tuning and model evaluation.
- Build and deploy end-to-end deep learning projects, from data preprocessing to model deployment, for real-world problem-solving.

Course Duration: The course duration is 40 sessions of 60 minutes each.

Course Content:

Module No.	Modules/Sub-Modules	No. of Sessions	% Weightage
I	**Practicals based on Neural Networks:** Neural Networks, Multilayer Perceptron, feedforward and back propagation algorithm, Digit classification using MNIST dataset.	**08**	**20%**
II	**Practicals based on** Autoencoders and Variational Autoencoders**,** Convolutional Autoencoders and Generative adversarial networks.	**08**	**20%**

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III	**Practicals based on** Convolution Neural Networks (CNN). Application: Digit classification, object detection	**08**	**20%**
IV	**Practicals based on Recurrent neural networks,** LSTM networks, neural sentiment analysis.	**08**	**20%**
V	**Practicals based on Transfer Learning:** Yolo, VGG, ResNet,etc.	**08**	**20%**

Course Outcomes: At the end of the course the students should be able to:

CO 1: Understand and implement basic neural networks in python programming.

CO 2: Implement autoencoders and GAN to real time applications

CO 3: Implement Convolution neural networks for solving problems.

CO 4: Implement recurrent neural networks to real world problems.

CO 5: Implement transfer learning networks to solve problems.

Teaching Scheme:

Teaching Scheme (Hours) : 40		
Course Credit: 4		
Theory	Tutorial	Practical
-	-	40

Examination Scheme:

Examination Scheme				
External	Internal	CEC		Total Marks
		Assignment	Attendance	
50	20	20	10	100

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Teaching Methods:

The following pedagogical tools will be used to teach this course:

1. Demonstration
2. Practical Implementation
3. Discussions
4. Assignments
5. Presentations

Supplementary Readings:

1. Ahmed Menshawy(2018). Deep Learning By Example. 1st Edition, Packt publications.
2. Luca Massaron, Alberto Boschetti, et.al. (2018). Tensorflow Deep Learning Projects. 1st Edition, Packt publications.
3. Josh Patterson, Adam Gibson (2018). Deep Learning A Practitioner's Approach. 2nd Edition, O'Reilly.

List of Journals / Periodicals / Magazines / Newspapers/ etc.:

The students can refer to the link below to get relevant information pertaining to the subject.

1. https://archive.nptel.ac.in/courses/106/106/106106184/
2. https://archive.nptel.ac.in/courses/106/105/106105215/
3. https://archive.nptel.ac.in/courses/106/106/106106201/

Session Plan:

Session	Topic
1	Practicals based on Neural Networks: Neural Networks,
2	Practicals based on Neural Networks: Neural Networks,
3	Multilayer Perceptron,
4	Multilayer Perceptron,
5	feedforward and back propagation algorithm,
6	feedforward and back propagation algorithm,
7	Digit classification using MNIST dataset.
8	Digit classification using MNIST dataset.
9	Practicals based on Autoencoders
10	Practicals based on Autoencoders
11	Variational Autoencoders
12	Variational Autoencoders
13	Convolutional Autoencoders
14	Convolutional Autoencoders
15	Generative adversarial networks.
16	Generative adversarial networks.
17	Practicals based on Convolution Neural Networks (CNN).
18	Practicals based on Convolution Neural Networks (CNN).
19	Practicals based on Convolution Neural Networks (CNN).
20	Practicals based on Convolution Neural Networks (CNN).
21	Application: Digit classification,
22	Application: Digit classification,
23	object detection
24	object detection
25	Practicals based on Recurrent neural networks,
26	Practicals based on Recurrent neural networks,
27	LSTM networks,
28	LSTM networks,
29	LSTM networks,
30	neural sentiment analysis.

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31	neural sentiment analysis.
32	neural sentiment analysis.
33	Practicals based on Transfer Learning: Yolo,
34	Practicals based on Transfer Learning: Yolo,
35	VGG,
36	VGG,
37	VGG,
38	ResNet,etc.
39	ResNet,etc.
40	ResNet,etc.

Suggested Practical Distribution:

The suggested distribution as per Bloom's taxonomy is as follows:

Distribution of practicals for course delivery and evaluation					
Remember	Understand	Apply	Analyse	Evaluate	Create
20%	20%	20%	10%	20%	10%

Instructional Method:

1. The course delivery method will depend upon the requirement of content and the needs of students.
2. The internal evaluation will be done on the basis of a practical exam and a continuous evaluation of students.

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